

Lab 05: EIGRP Configuration & Verification

CCNA Lab Portfolio | Cisco Packet Tracer Documentation

Author: Luigi

Date: August 30, 2026

Program: BS Information Technology | Protocol: EIGRP | Previous Lab: Lab 04 – OSPFv2

1. Objectives

- Explain what EIGRP is and how it differs from OSPF (configured in Lab 04).
- Enable classic EIGRP on Cisco IOS routers using an Autonomous System (AS) number.
- Advertise networks correctly using the network command and wildcard masks.
- Prevent unnecessary EIGRP hellos on LAN-facing interfaces using passive-interface.
- Disable automatic summarization using no auto-summary.
- Verify neighbor adjacencies, the topology table, and the routing table.
- Troubleshoot common EIGRP adjacency problems.

2. What is EIGRP?

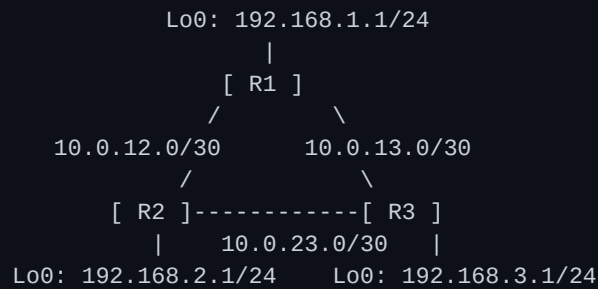
EIGRP is a Cisco-proprietary (now partially open, RFC 7868) advanced distance-vector routing protocol. Unlike OSPF, which is a link-state protocol that floods full topology information and runs Dijkstra's SPF algorithm, EIGRP uses the **Diffusing Update Algorithm (DUAL)** to calculate loop-free paths, keeping a Successor, Feasible Successor, and full topology table per network.

Characteristic	EIGRP
Administrative Distance	90 (internal), 170 (external)
Metric basis	Composite: Bandwidth + Delay by default
Convergence	Very fast — DUAL pre-computes backup paths
Neighbor discovery	Multicast Hello to 224.0.0.10, every 5s
Design unit	Single Autonomous System number (no areas)

3. Repository Files

File	Purpose
00-Overview-and-Author.md	Objectives, background, author info
01-Topology-and-Addressing.md	Network diagram and IP addressing table
02-CLI-Commands-and-Configuration.md	Full CLI configuration with EIGRP command explanations
03-Verification-and-Testing.md	Verification commands, expected output, troubleshooting

4. Topology Diagram



Three routers in a triangle so EIGRP forms adjacencies over more than one link, plus a loopback on each router simulating a LAN behind it. A triangle (vs. a straight line) also allows testing Feasible Successors by shutting down a link.

5. IP Addressing Table

Device	Interface	IP Address	Mask	Wildcard
R1	Gig0/0/0 (to R2)	10.0.12.1	255.255.255.252	0.0.0.3
R1	Gig0/0/1 (to R3)	10.0.13.1	255.255.255.252	0.0.0.3
R1	Loopback0	192.168.1.1	255.255.255.0	0.0.0.255
R2	Gig0/0/0 (to R1)	10.0.12.2	255.255.255.252	0.0.0.3
R2	Gig0/0/1 (to R3)	10.0.23.1	255.255.255.252	0.0.0.3
R2	Loopback0	192.168.2.1	255.255.255.0	0.0.0.255
R3	Gig0/0/0 (to R1)	10.0.13.2	255.255.255.252	0.0.0.3
R3	Gig0/0/1 (to R2)	10.0.23.2	255.255.255.252	0.0.0.3
R3	Loopback0	192.168.3.1	255.255.255.0	0.0.0.255

EIGRP Autonomous System number used throughout this lab: **AS 100**. This value only needs to match between routers in the same domain.

6. EIGRP-Specific Commands

Command	What it does
router eigrp <AS>	Starts the EIGRP process under Autonomous System <AS>. Must match on every adjacent router.
network <addr> <wildcard>	Advertises a locally connected interface. Uses a wildcard mask (inverse of subnet mask), not a regular mask.
no auto-summary	Disables automatic summarization at classful boundaries — required for discontinuous designs.
passive-interface <int>	Stops Hellos on an interface (e.g. a LAN) while still advertising its network.
eigrp router-id <id>	Manually sets the EIGRP router ID instead of relying on auto-selection from loopback/interface IPs.
bandwidth <kbps>	Sets the interface bandwidth value used in EIGRP's metric (label only, not physical speed).
delay <10s of μs>	Sets interface delay, the other half of EIGRP's default composite metric.
show ip eigrp neighbors	Lists routers with a formed EIGRP adjacency — the first command to check.
show ip eigrp topology	Shows DUAL's Successor (installed) and Feasible Successor (backup) per network.
show ip protocols	Summarizes the running process: AS number, timers, advertised networks.
show ip route eigrp	Filters the routing table to EIGRP-learned routes only (marked D).

7. Configuration — R1

```
enable
configure terminal
hostname R1

interface GigabitEthernet0/0/0
description Link-to-R2
ip address 10.0.12.1 255.255.255.252
no shutdown
exit

interface GigabitEthernet0/0/1
description Link-to-R3
ip address 10.0.13.1 255.255.255.252
no shutdown
exit

interface Loopback0
description R1-LAN-simulated
ip address 192.168.1.1 255.255.255.0
exit
```

```
router eigrp 100
  eigrp router-id 1.1.1.1
  network 10.0.12.0 0.0.0.3
  network 10.0.13.0 0.0.0.3
  network 192.168.1.0 0.0.0.255
  passive-interface Loopback0
  no auto-summary
exit

end
write memory
```

8. Configuration — R2

```
enable
configure terminal
hostname R2

interface GigabitEthernet0/0/0
  description Link-to-R1
  ip address 10.0.12.2 255.255.255.252
  no shutdown
exit

interface GigabitEthernet0/0/1
  description Link-to-R3
  ip address 10.0.23.1 255.255.255.252
  no shutdown
exit

interface Loopback0
  description R2-LAN-simulated
  ip address 192.168.2.1 255.255.255.0
exit

router eigrp 100
  eigrp router-id 2.2.2.2
  network 10.0.12.0 0.0.0.3
  network 10.0.23.0 0.0.0.3
  network 192.168.2.0 0.0.0.255
  passive-interface Loopback0
  no auto-summary
exit

end
write memory
```

9. Configuration — R3

```
enable
configure terminal
hostname R3

interface GigabitEthernet0/0/0
description Link-to-R1
ip address 10.0.13.2 255.255.255.252
no shutdown
exit

interface GigabitEthernet0/0/1
description Link-to-R2
ip address 10.0.23.2 255.255.255.252
no shutdown
exit

interface Loopback0
description R3-LAN-simulated
ip address 192.168.3.1 255.255.255.0
exit

router eigrp 100
eigrp router-id 3.3.3.3
network 10.0.13.0 0.0.0.3
network 10.0.23.0 0.0.0.3
network 192.168.3.0 0.0.0.255
passive-interface Loopback0
no auto-summary
exit

end
write memory
```

10. Step-by-Step Activity Flow

- 1 Cable the three routers into the triangle topology.
- 2 Configure hostnames and interface IPs on R1, R2, R3 (no EIGRP yet); no shutdown every interface.
- 3 Confirm show ip interface brief shows up/up everywhere before touching EIGRP.
- 4 Enable EIGRP AS 100 on R1 with its network statements, then move to R2.
- 5 Enable EIGRP on R2; check show ip eigrp neighbors before continuing.
- 6 Repeat for R3 — all three routers should now see two neighbors each.
- 7 Confirm no auto-summary and passive-interface Loopback0 are present on all three.
- 8 Run the full verification and troubleshooting checklist (Section 12).

11. Verify Neighbor Adjacencies

```
R1# show ip eigrp neighbors
```

```
EIGRP-IPv4 Neighbors for AS(100)
```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RT0	Q Cnt	Seq Num
1	10.0.13.2	Gi0/0/1	14	00:02:10	40	200	0	12
0	10.0.12.2	Gi0/0/0	12	00:03:45	35	200	0	15

Two neighbors expected on R1 (R2 and R3). An empty list means adjacency is not forming — see the troubleshooting table below.

12. Verify Topology & Routing Tables

- **show ip eigrp topology** — look for state P (Passive/healthy) vs A (Active/recalculating); note the Successor and any Feasible Successor.
- **show ip route eigrp** — R1 should show D routes for 192.168.2.0/24, 192.168.3.0/24, and 10.0.23.0/30.
- **show ip protocols** — confirm the AS number, network statements, and 'Automatic network summarization is not in effect'.

13. End-to-End Connectivity Test

```
R1# ping 192.168.2.1
```

```
R1# ping 192.168.3.1
```

14. Troubleshooting Reference

Symptom	Likely Cause	Fix
Neighbors list empty between directly connected routers	AS number mismatch	Match router eigrp <AS> on both sides
Neighbors won't form, interfaces up/up	Subnet mismatch	Recheck IPs/masks vs. addressing table
Neighbors won't form despite matching AS/subnet	K-value mismatch	Compare K1-K5 via show ip protocols on both sides
Neighbor flaps up/down	Timer mismatch or flapping link	Check show interfaces and show ip eigrp interfaces detail
Route missing from routing table	Missing/incorrect network statement	Re-check wildcard mask on the network line
Routes look summarized (e.g. only /16 shown)	auto-summary still enabled	Add no auto-summary under router eigrp

Symptom	Likely Cause	Fix
Unexpected adjacency on LAN interface	Missing passive-interface	Add passive-interface Loopback0

15. Lab Completion Checklist

- All three routers show 2 neighbors each in show ip eigrp neighbors
- All six subnets appear in every router's show ip route eigrp
- show ip protocols confirms no auto-summary is active on all routers
- End-to-end pings between all three loopback addresses succeed
- Screenshots of show ip eigrp neighbors and show ip route eigrp saved for the write-up

End of Lab 05 documentation.